

Zidane Zahro Novaro

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Highly analytical Universitas Indonesia graduates with a proven track record of driving **operational excellence and continuous improvement** across fast-paced environments. Demonstrates multifaceted experience in **data-driven problem solving, process optimization, and root cause analysis** utilizing Lean & DMAIC/PDCA/FMEA frameworks. I have strong interest in diverse industries including **smelter, mineral processing, laboratories, manufacturing, mining, oil & gas, banking, consulting, retail, education, supply chain/logistics, FMCG, tobacco, Agroculture, and others**. Equipped with robust **integrated risk management capabilities**, backed by **BNSP General OHS and ISO 9001, 14001, 45001** certifications to ensure strict operational compliance. A proven strategic leader who has successfully managed **55-person cross-functional teams** and **divisional budgets**, showcasing strong financial control and adaptability to contribute transferable expertise in **analytics & compliance** across Functional sectors.

- **Courses / Certifications / Skills:** English Proficiency test (EPT)/Toefl by LIB UI : Score 503, BNSP Certified General Occupational Safety & Health Expert (Competent), and advanced software tools including Excel, Origin, SolidWorks, Word/PPT, and multimedia editing software.

EDUCATION

Bachelor of Metallurgical & Materials Engineering, part of Mechanical Engineering Faculty, Universitas Indonesia (2021/08 – 2025/07)

- GPA: 3.53 / 4.00
- 1st Place in the 2024 Chromatic UI Essay Competition on Corrosion and Protection Materials, top 7 finalists of the Student Innovation Forum on Critical Minerals at MnMs UI 2024, top 10 finalists of the 2023 DEF Undip Business Plan Competition, and various "Best Staff" awards at IMMT FT UI, Nano Rsc FTUI and GP Tech SC UI,.

Math and Science, SMA Negeri 1 Kota Bekasi (2018/08 – 2021/06)

- Average Grade : 90
- Ranked 24th in the average report card of science majors at SMAN 1 Bekasi for **SNMPTN**

CORE COMPETENCE

1. Operational Excellence & Continuous Improvement

Applied **Lean Manufacturing, Kaizen, and TPS** at PT Pakoakuina to monitor **OEE and cycle times**, and evaluated **TPM** at PT Astra Honda Motor to consistently achieve plant **QCDSM targets**.

2. Data-Driven Analysis & Problem Solving

Proficient in processing **700+ RBI data entries** and BRIN composite research, utilizing **Fishbone and DMAIC frameworks** to diagnose root causes and deliver **actionable insights** for industrial optimization.

3. Integrated Risk Management, EHS & Compliance

Ensures **operational compliance** through **BNSP General OHS and ISO 9001, 14001, 45001** certifications, backed by expertise in developing **feasibility studies and RKAB documents** aligned with regulatory standards.

4. Strategic Project Leadership & Financial Control

Led a **55-person cross-functional team** as Project Officer for Metal Seraga and managed **divisional budgets** as Head of Treasury for Nano RSC, complemented by winning **1st Place in a National Essay Competition**.

WORK EXPERIENCES

Management Trainee - SLDP Production, PT Prakarsa Alam Segar (Wings Group) (2026/04 - Present)

- Lead shopfloor operations and production maintenance to optimize factory efficiency, ensuring strict compliance with safety, quality, and manufacturing standards.
- Collaborate with cross-functional teams in trouble-shooting technical issues and managing team performance to drive continuous process improvement.

Independent Business, Projects & Digital Services (2022/07 - Present)

- Worked on other people's assignments, papers, etc. and competed for high rankings in Mobile Legends.
- Selling premium app accounts like Netflix, Canva, etc.

Pic Non-Lab Polymer Laboratory Assistant, Lab Polymer DTMM FT UI**(2023/06- 2023/12)**

- Assisted 2-6 researchers and practitioners in conducting experiments and research related to polymers
- Maintain the laboratory environment & provide technical support and training.

Assistant lecturer for Calculus 1, Electrical Physics Mechanics, Waves, & Optics, and Introductory Engineering Materials, FT UI**(2022/08 – 2024/12)**

- Responsible for creating and checking assignments, quizzes, and exam materials; preparing solution keys; and providing tutoring sessions before exams according to student needs.
- Assisted in three different courses across multiple semesters, supporting a total of 162 students in five classes (WFA) and contributed to improving student understanding and academic performance through structured problem-solving guidance and personalized feedback

INTERNSHIP EXPERIENCES**Melting Casting Production Engineer Intern, PT Pakoakuina 2W (PAKO GROUP)** **(2025/12 – 2026/04)**

- Applied core manufacturing principles by learning and implementing Pako core values, safety culture, whistleblowing system, risk management, shopfloor management, kaizen, and 8-step improvement methodologies within daily operations.
- Supported melting and casting operations and heat treatment processes through safety, quality, productivity, and cost monitoring, applying TPS tools such as OEE, OK ratio, monitoring actual/cycle time, SOP/WI, TSKK, and Job ALO observation (Yamazaki Chart) to identify abnormalities and improvement opportunities.
- Executed individual improvement projects to improve production efficiency and reduce operational cost in Melting Casting 2W division, using structured 8-Step continuous improvement tools to drive measurable operational gains.

Metallurgist - MCC Intern, PT Koninis Fajar Mineral (28 Ayoda Group)**(2025/05 – 2025/11)**

- Assisted in preparing RKAB documentation and feasibility studies in compliance with ESDM Regulation No. 1806/2018, processing and maintaining on-site IUP data, and compiling comprehensive reports for the head office.
- Performed comparative analysis of nickel processing routes (HPAL vs RKEF), assessed silica sand purification for photovoltaic glass specifications, and evaluated vanadium and scandium recovery rates and market viability — providing data-driven insights for strategic planning.

Failure Analysis Engineer, CPMFA UI**(2024/09 – 2025/02)**

- Completed failure analysis on national monument statues in collaboration with Ministry of Culture and analyzed stainless steel SS 316L tube leakage for hydrocarbon gas flow identifying pitting corrosion and recommending coating and material specification improvements
- Diagnosed root cause of drill pipe failure using Fishbone Diagram and DMAIC framework for PT Sucofindo and Pertamina Hulu Rokan and coiled tubing failure for PT Schlumberger using Fishbone Diagram and DMAIC framework

Quality Control Engineer intern Quality & HSE division, PT Bakrie Pipes Industrie **(2024/11 – 2024/12)**

- Conducted failure analysis on stainless steel structures of national monument statues for Plate Lamination to reduce A1-class defects on API 5L L390M (X56M) PSL2 conductor pipes for PT Pertamina Hulu Mahakam.
- Collaborated with QHSE team to identify inspection inefficiencies and improvement opportunities in inspection setup, improve couplant handling, and reduce calibration frequency for KT-24 HFW pipe production.

Production and Process Engineer Casting wheel Intern, PT Astra Honda Moto**(2024/01 – 2024/03)**

- Analyzed production capacity and TPM performance to ensure optimal QCDSM outcomes.
- Working & Investigated on the project “The Effect Of Temperature & Cycle Time In Heat Treatment Of Silicon Aluminium Alloy Castings On Mechanical Properties Of Wheel Front K2SA”(ANALYZE 2 VARIABLES namely temperature 510 ° C, 515 ° C, 520 ° C, 525 ° C,, 530 ° C and holding time 25, 27, and 28 mnts). Performed microstructure analysis, hardness testing, and metallographic sample preparation and evaluation

Risk Based Inspection pipeline Intern, PT. Integracor Prima Nusantara**(2023/07 – 2023/09)**

- Processed and managed 700+ RBI pipeline inspection data entries (wth, utm, pimps).
- Conducted Risk-Based Inspection (RBI) pipeline analysis (2021) – Pertamina Hulu Rokan & Risk based inspection of pipeline (prior to 2021) – Chevron Pacific Indonesia Responsible for : Conduct Risk-Based Inspection (RBI)-Pipeline at Blok Rokan, MINAS Field as part of Asset Integrity Management System for Chevron and Pertamina.

RESEARCH EXPERIENCES

Undergraduate Research | BRIN RI, Cibinong & CSWM UI

(2024/06 – 2024/12)

(2025/04 – 2025/05)

- Conducted research and material characterization of Wood Polymer Composites using reinforced recycled polyethylene (r-PE) and Oil Palm Empty Fruit Bunch (OPEFB/TKKS), applying plasma & alkalization treatments for comparative fiber and mechanical performance study. Operated and optimized advanced laboratory equipment (extruders, pelletizers, mini injection moulding, pulverizers, chopper, shredder, grinder) to process oil palm biomass into fiber/flour and formulate composite materials for testing.
- Assisted in data analysis, manuscript drafting, and graph preparation for WPC/PMC research using r-PE and OPEFB (5– 20%) with various treatments, comparing fiber and mechanical properties to Rice Husk and Teak for a Q2 journal submission (Taylor & Francis).

ORGANIZATIONAL & COMMITTEE EXPERIENCES

Metal Seraga Project Officer Committee

(2022/08 – 2022/12)

- Led a committee of 55 people to organize an art, sports, and science competition between active batches at DTMM FTUI. Made a project plan for the next 6 months, worked together, think critically, exchanged opinions, executed and accommodated the needs of the community for the event to run smoothly.
- Successfully executed the event from planning to completion with the innovation of new sports and new division

Head of Treasury Division, Nano RSc FTUI

(2023/01 – 2023/12)

- Led and supervised a treasury team of 5 staff, plan, organize and supervise the implementation of 6 divisional work programs and activities, including: profit and loss agreements, cash flow, Annual Activity Plan and Budget, management jacket, membership fees, financial reports.
- Analyze financial data, make projections, and provide strategic recommendations for management decision making. As well as identifying cost efficiency opportunities and optimizing the allocation of financial resources.

Deputy Head of Media and Publication Division, GP Tech SC UI 2023

(2023/01 – 2023/12)

- Lead and coordinate the media and publication team consisting of 5 staff members. Plan, organize, and oversee the implementation of 7 work programs and division activities, including: Social Media Instagram Branding, Online magazine GP-Zine, Documentations for Company visit to SLB, Pengabdian Masyarakat, Seminar waste plastic management seminar, Polytalk Podcast & make Video Profile.
- successfully to create the highest number of video reels with 6,122 views regarding the polytalk eps 2 trailers on Instagram

Staff Human Resources Management and organization, UKM UI Esport Club

(2022/02 – 2022/12)

- Improve the quality of human resources, other fields, and internal bonding, which includes registering and wishing members happy birthdays, holding monthly meetings, and successfully hosting a PUBG Mobile x Mobile Legends bonding event with gaming and conversations at a coffee shop.

COURSE & CERTIFICATION

→ Published by Makin Ahli, Makin Ahli pangan Official

(2026- Present)

- **Food Safety (Good Manufacturing Practices, HACCP, ISO 22000:2018, FSSC 22000 V 7.0)**

- **Quality Management System (ISO 9001:2015) in Food Industry**

→ Published by LSP Katigapass, PT Mitra Dinamis Utama (Midiatama), BNSP

(2025- 2028)

- **Certified General Occupational Safety & Health Expert (Competent)**

→ Published by Talenthub by Kemenaker RI

(2025- Present)

- **Talenthub batch 19, Mandarin Language for Professionals**

→ Published by Nestle (2025- Present)

- **Nestle E-learning 2025 | Sustainability Topics**

→ Published by PT. Yura Prima Solusindo

(2024- Present)

- **ISO 14001:2015 Environmental Management Systems Certified**

- **ISO 45001:2018 Occupational Health and Safety Management Systems (K3) Certified**

- **ISO 9001:2015 Quality Management System Certified**

→ Published by Lembaga Bahasa Internasional, Fakultas Ilmu Pengetahuan Budaya, Universitas Indonesia (LBI FIB UI) with the document number 18761/UN2.F7.LBI/PDP.00.07.01/2021

(2021- Present)

- **The English Proficiency Test (EPT) : 503 (equivalent to standardized TOEFL scores Section I**
(Listening Comprehension): Raw 36, Converted 53;
Section II (Structure and Written Expression): Raw 27, Converted 50;

VOLUNTEERS EXPERIENCES

- Mentor OKK UI 2023 (2023/06 – 2023/08)
- Jakarta sadar sampah x BEM UI 2022 Volunteer (2022/07 – 2022/09)

HONORS & AWARDS

- **1st** Essay Competition about Corrosion and Protection Material at Cromatic UI 2024.
- **Top 7** finalist teams Student Innovation Forum Competition about critical Minerals at MnMs UI 2024.
- **Top 10** Finalist Proposal Business Plan Competition, DEF Undip 2023
- **Best staff treasury** triwulan 1 at Nano RSc FT UI 2022 (HRE : 3.77), **Best staff relation** Triwulan 1 at GP Tech SC UI 2022 (HRE : 3.79) & **Best Staff** Of The Term Quarter 1 at **Sport and Arts Division** in IMMT FTUI 2022 (HRE: 3.70)
- **2nd Place** – Outstanding Male Student Competition, Mustikajaya Subdistrict Scale, Bekasi City.

SKILLS

- **Language:** Indonesia (Native), English (EPT 503) , Mandarin (intermediate), Japanese & Korean (Beginner)
- **Software:** Origin, Microsoft office (word, power point, Excel, etc), Google workspace (g-docs, g-slide, g-spreadsheet), solidworks, matlab, artstep, obs studio, canva, filmora, adobe premier pro, capcut, lightroom.
- **Others:** Public Speaking, Project Planning Management, Critical Thinking, collaboration and teamwork, Creative, act creatively and think visionarily, strong desire to learn new things, Photography, Design Photo, Editing Video & Writing literature journal/article.

TITLE BACHELOR THESIS

- "Comparative Analyze study of Modification of Oil Palm Empty Bunches with Plasma Treatment and Alkalization as Reinforcement for Sustainability Wood Plastic Composite Applications" about investigates the effect of modifying palm oil empty bunches (OPEFB) through plasma treatment and alkalization as reinforcement in sustainable wood-plastic composite applications, aiming to enhance the mechanical and functional properties of the material

Publication

- Z. Z. Novaro, "Modifikasi tandan kosong kelapa sawit dengan perlakuan plasma dan alkalisasi sebagai penguat untuk aplikasi komposit plastik kayu," Skripsi, Dept. Teknik Metalurgi dan Material, Univ. Indonesia, Depok, Indonesia, 2025. <https://lib.ui.ac.id/detail?id=9999920564755&lokasi=lokal>

References

- Yusuf Nafi Farhan, S.T. (Management Trainee at Wuling Motors Indonesia)
Relationship: Close friend from college (Email: yusuf.nafi@gmail.com)
- Raden Nafian (Section Head – New Model & Packing Export, PT Astra Honda Motor)
Relationship: Internship/Practical Work Mentor, (Email: raden.n.priatmojo@astra-honda.com)
- Muhammad Faiq Zuhdi, EV Battery Engineer, PT Hyundai LG Indonesia Green Power (HLI)
Relationship: University Senior Mentor, (Email: muhammad.faiq@ui.ac.id)
- Marcell Twi, (Head of Engineering Section, PT Integricor Prima Nusantara)
Relationship: Internship Supervisor / Mentor. (Email: marcell.twi@integricorprima.com).